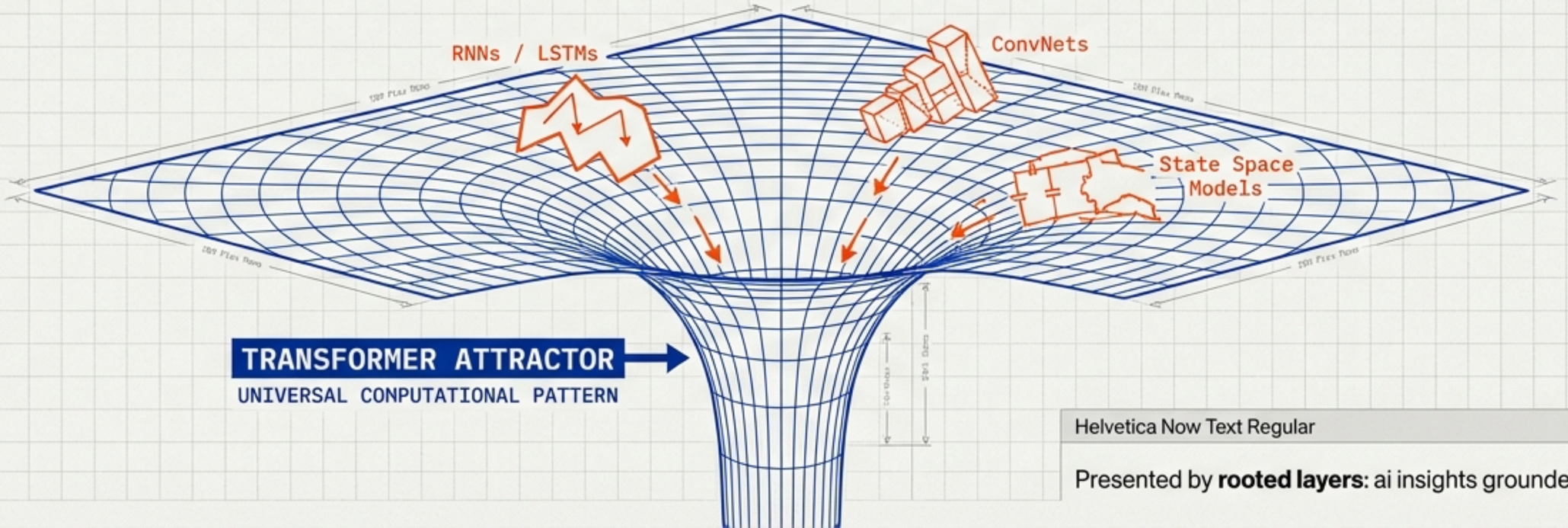
THE TRANSFORMER ATTRACTOR

Why every alternative architecture converges toward the the same computational pattern—dies trying.



Helvetica Now Text Regular IBM Plex Mono

Presented by **rooted layers**: ai insights grounded on research.

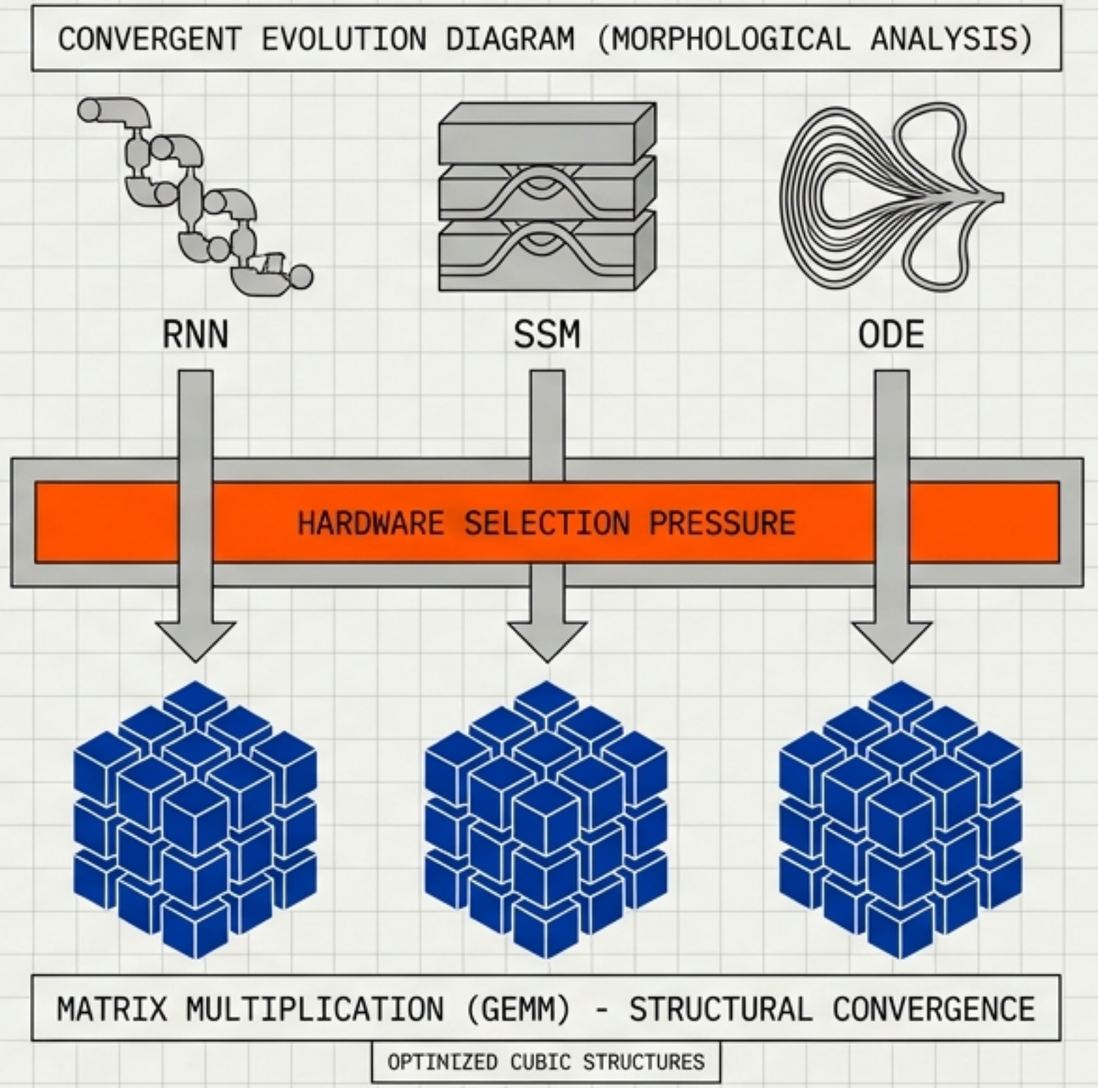
NotebookLM

Innovation is no longer just technical—it is structural, economic, and institutional.

Transformer dominance is a result of **convergent evolution**. Alternatives survive only by “wearing the Transformer as a disguise” (optimizing for GEMMs).

A **Stable Equilibrium** has formed between GPU silicon and Transformer architecture.

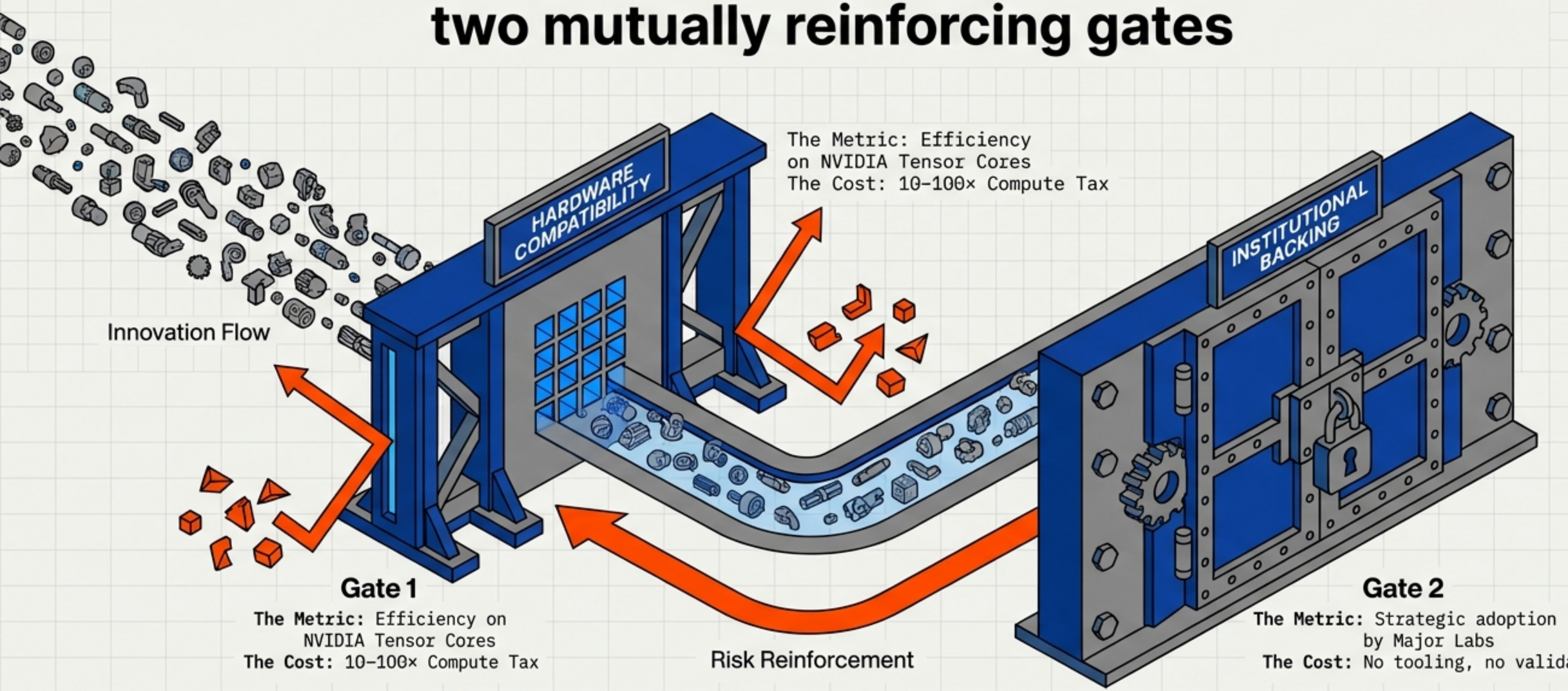
The next breakthrough will not replace the Transformer; it will happen *within* the Transformer’s constraints.



“The architecture changed to fit the hardware. The hardware did not budge.”

NotebookLM

Escaping the attractor requires clearing two mutually reinforcing gates



Gate 1

The Metric: Efficiency on NVIDIA Tensor Cores
The Cost: 10–100× Compute Tax

Gate 2

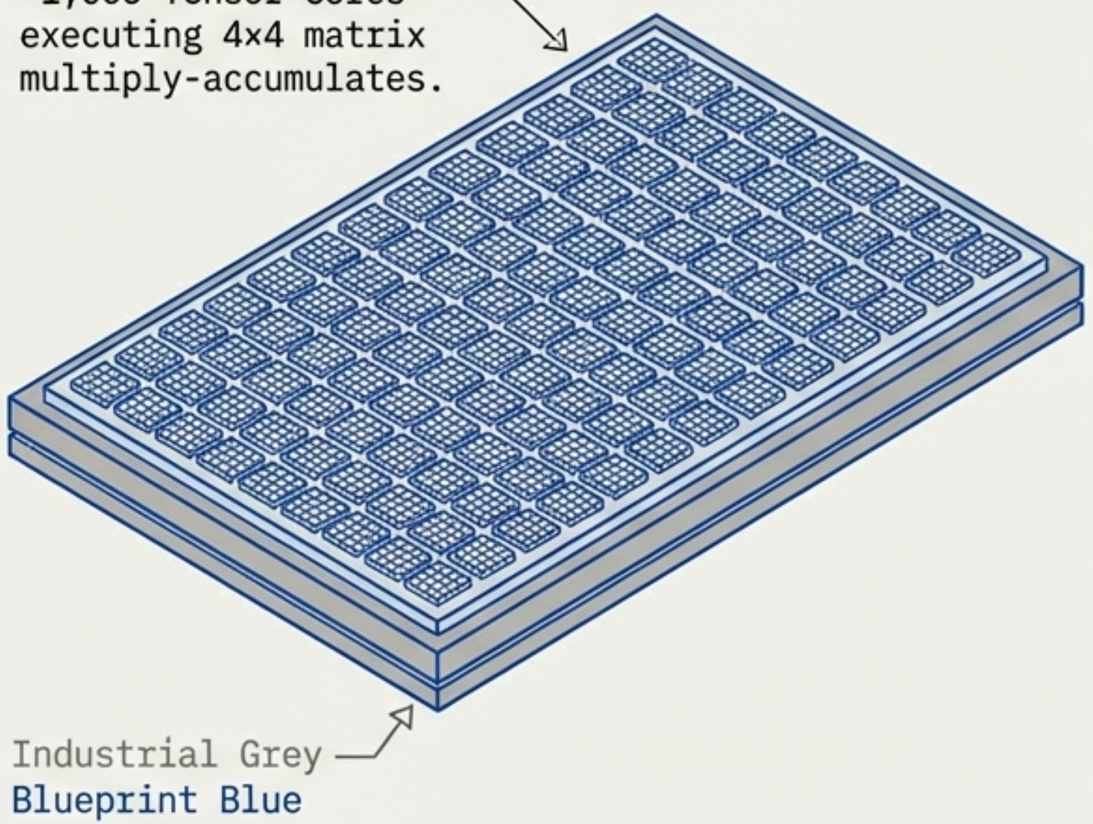
The Metric: Strategic adoption by Major Labs
The Cost: No tooling, no validation.

At frontier scale, breaking out requires changing both simultaneously—a coordination problem no single actor can solve.

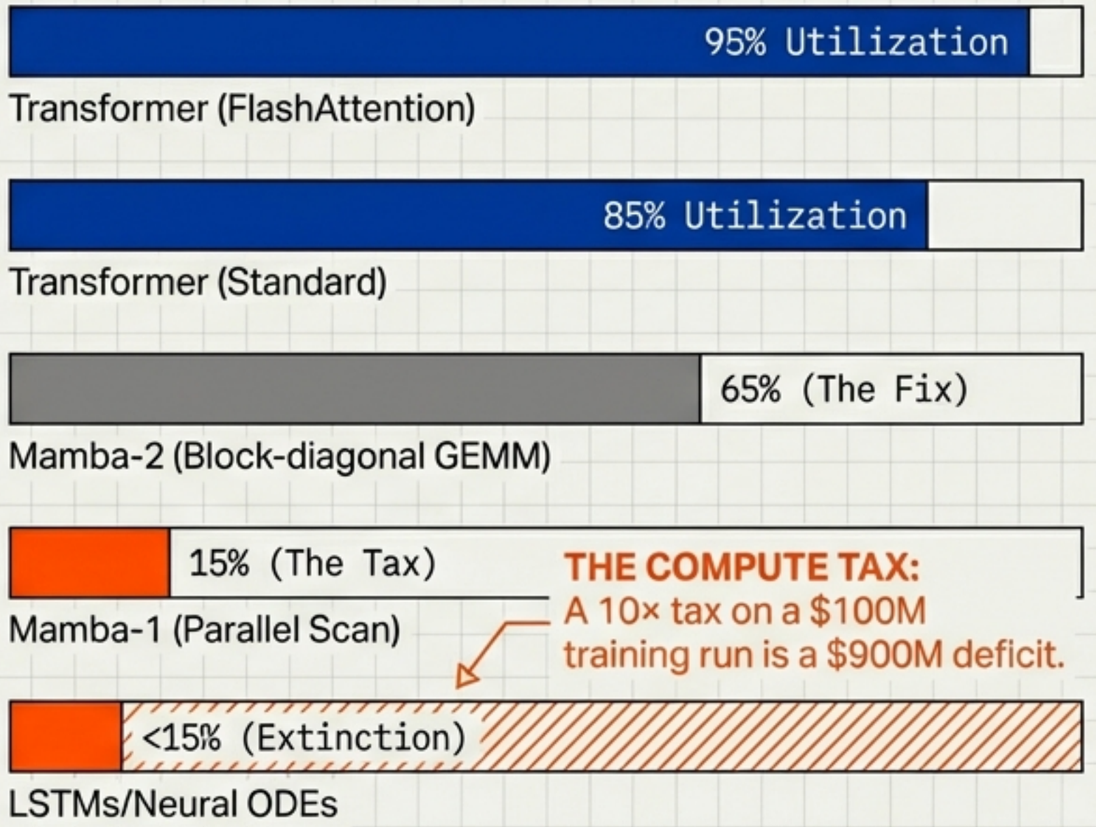
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The H100 is not a neutral processor; it is a GEMM optimization engine.

H100 CAPACITY



TENSOR CORE UTILIZATION BY ARCHITECTURE



NotebookLM

Sequential dependencies create pipeline stalls that GPUs cannot forgive.

DATA PARALLELISM (Transformer)
Batched Matrix Multiplies:
Independent Tiles



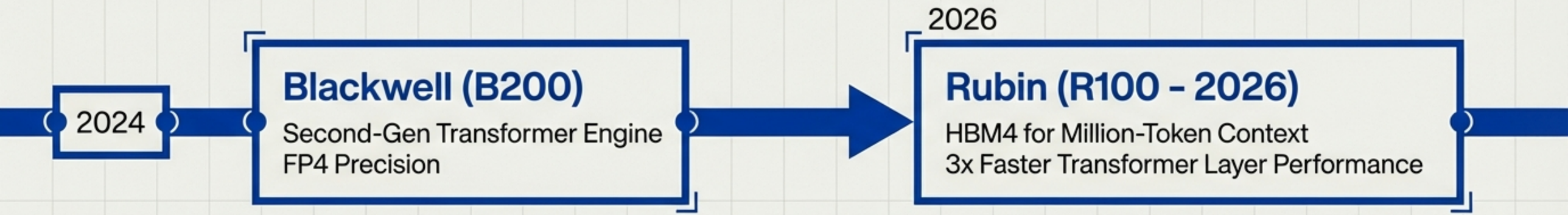
DATA DEPENDENCY (Scan/RNN)
Parallel Scan: Token B needs Token A



LSTMs/GRUs: Training cannot parallelize.
Neural ODEs: Iterative solvers allow elegant math but catastrophic economics.
KANs: Irregular computations starve the massive Tensor Cores.

NotebookLM

NVIDIA's roadmap is a $30B directional bet against non-GEMM architectures.



THE VOID (Unsupported Primitives)

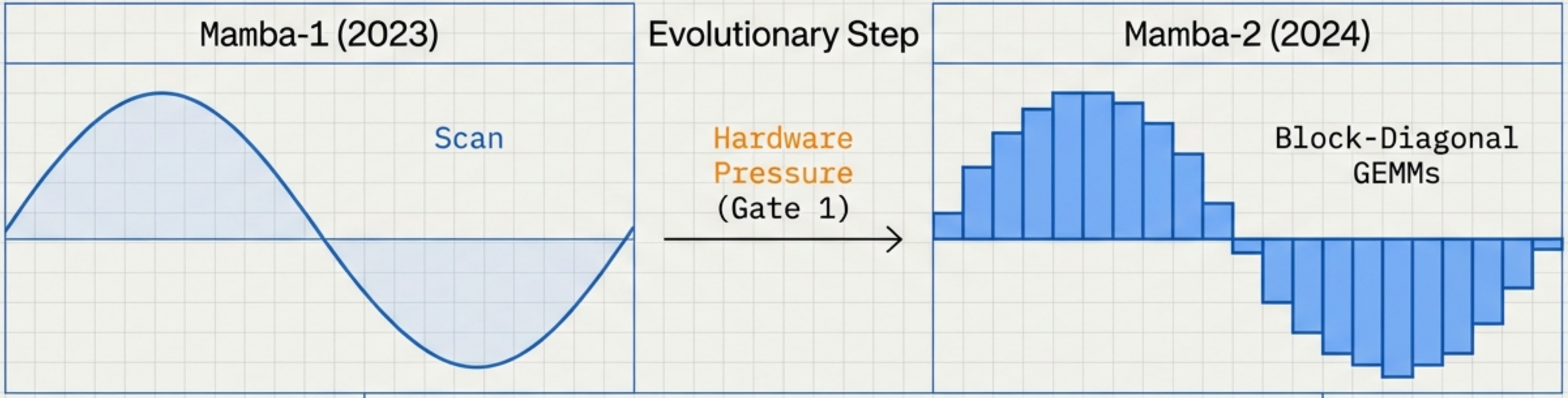
- No Parallel Scan Accelerators
- No Recurrence Units
- No Irregular Memory Support

This is **Directional Investment.** It compounds the advantage of GEMM-aligned architectures through 2028.

NotebookLM

Case Study I: Mamba-2 survived by rewriting its math to look like a Transformer.

Survival by Conformity



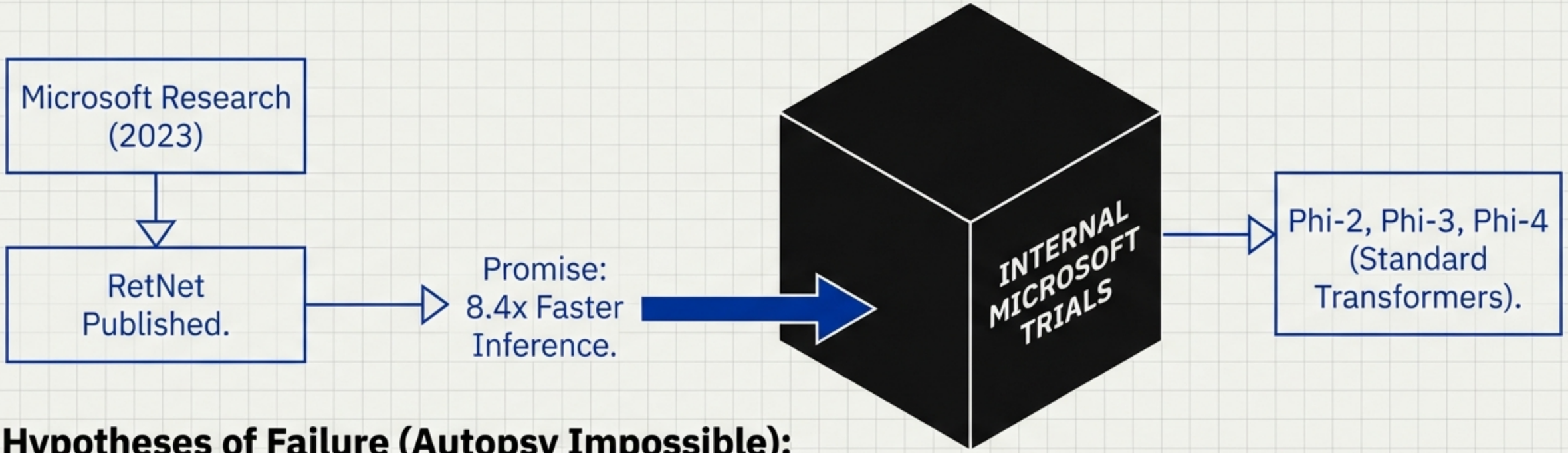

```
Speedup: 2-3x.
Gate 1 Status: Improved (60-70% Utilization).
Gate 2 Status: FAILED (No frontier deployment >30B).
```

"We restrict the SSM structure to allow efficient computation via matrix multiplications." — Mamba-2 Authors

NotebookLM

Case Study II: RetNet proves that the trap is opaque

The Diagnostic Blind Spot



Hypotheses of Failure (Autopsy Impossible):

1. **Hidden Hardware Friction**: Bottlenecks at 60+ layers?
2. **Quality Degradation**: Fixed-state limits reasoning?
3. **Risk Aversion**: Refusal to pay ecosystem tax?

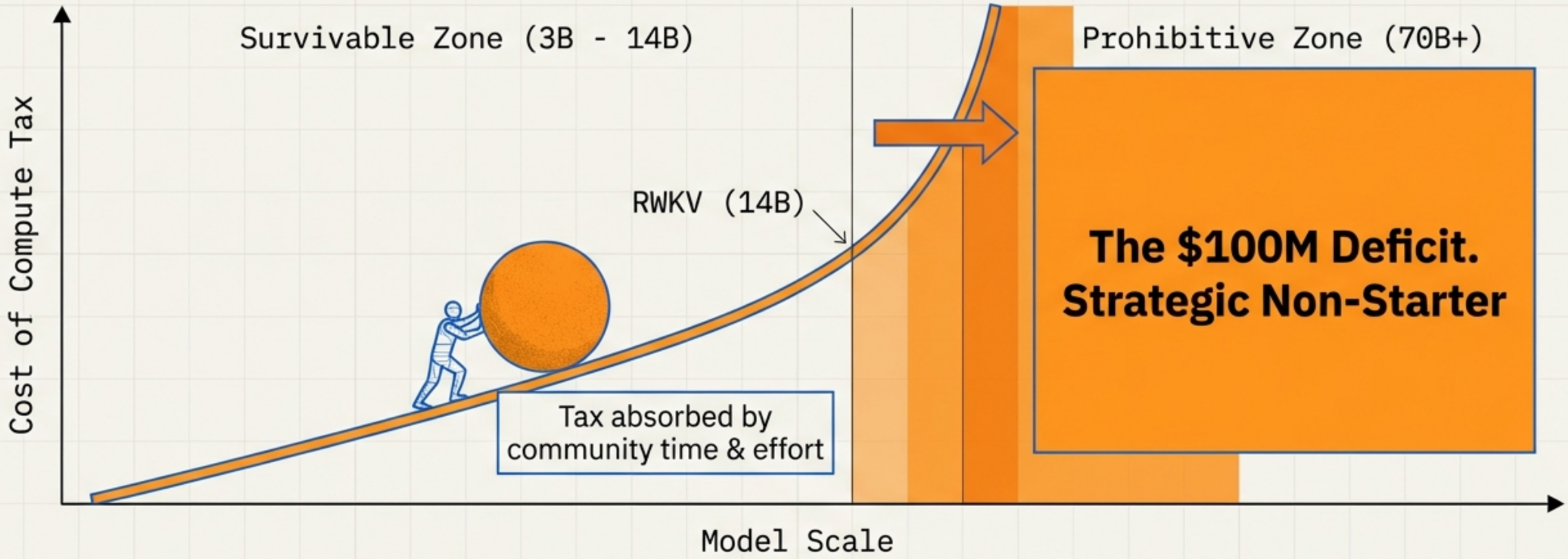
Without public failure data, we cannot learn why it failed. The trap remains invisible.

NotebookLM

Computational Brutalism meets High-End Editorial

Case Study III: The Attractor acts as a gradient field

RWKV and the Scale Dependency



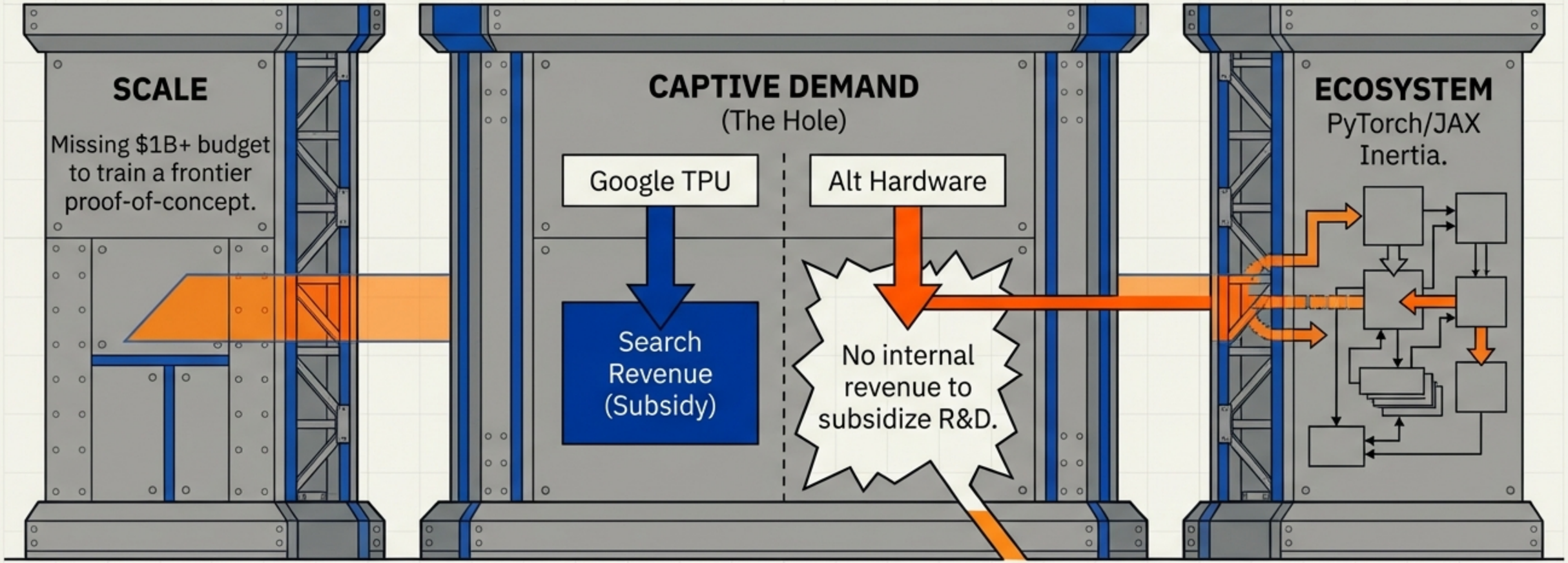
Frontier scale requires BOTH gates.
Community support is not enough for 100B+ models.

NotebookLM

Computational Brutalism meets High-End Editorial

Alternative hardware faces a ‘Captive Demand’ hole.

Barriers to Entry for Groq/Cerebras.



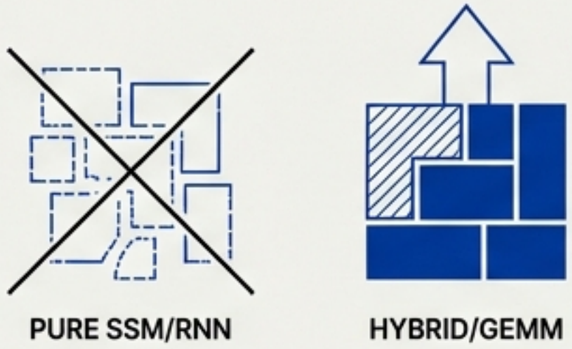
<10% market share by 2028 without a massive lab partnership.

NotebookLM

Signs of Lock-In: Predictions for 2027–2028

01 // No Pure Non-GEMM Frontier Model

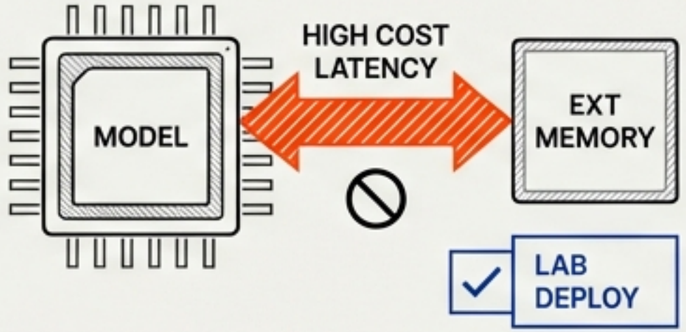
No top-tier LLM (>30B) will be a pure SSM/RNN. All winners will be Hybrids or GEMM-aligned.



IBM Plex Mono
SCALE: >30B // ARCH: HYBRID/GEMM-ALIGNED // STATUS: PROJECTED

02 // Titans Remain Research-Only

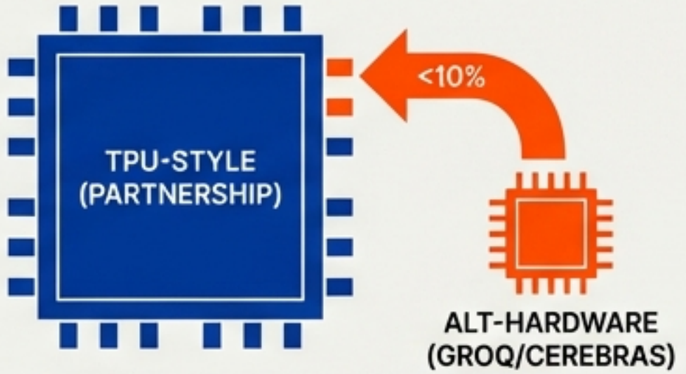
External memory architectures will **not** ship as flagship models due to serving topology costs.



IBM Plex Mono
MEM: EXTERNAL // COST: SERVING TOPOLOGY // DEPLOY: RESEARCH ONLY

03 // Alt-Hardware Stays Niche

Vendors like Groq/Cerebras will **not** exceed 10% market share without an exclusive "TPU-style" partnership.



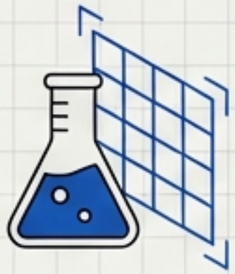
IBM Plex Mono
VENDOR: ALT-HARDWARE // SHARE: <10% // PATH: PARTNERSHIP

<10% market share by 2028 without a massive lab partnership.

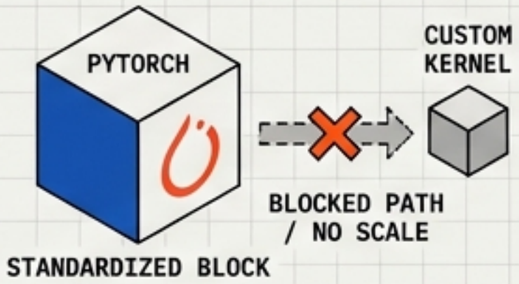
NotebookLM

Navigating the Lock-In: Heuristics for Practitioners.

RESEARCHERS



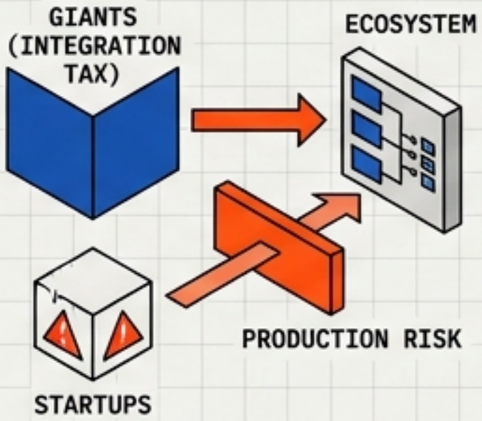
Don't fight the lock-in blindly. If it requires custom kernels not upstream in PyTorch, it will not scale.



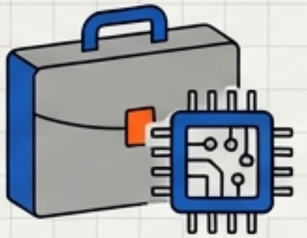
STARTUPS



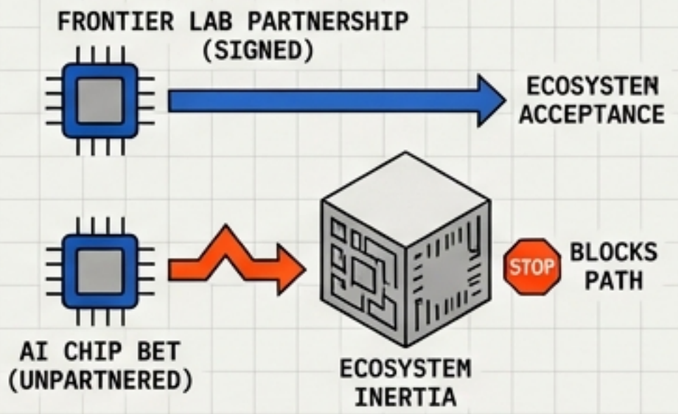
Wait for giants to pay the integration tax. If it's not in PyTorch, it's too risky for production.



INVESTORS



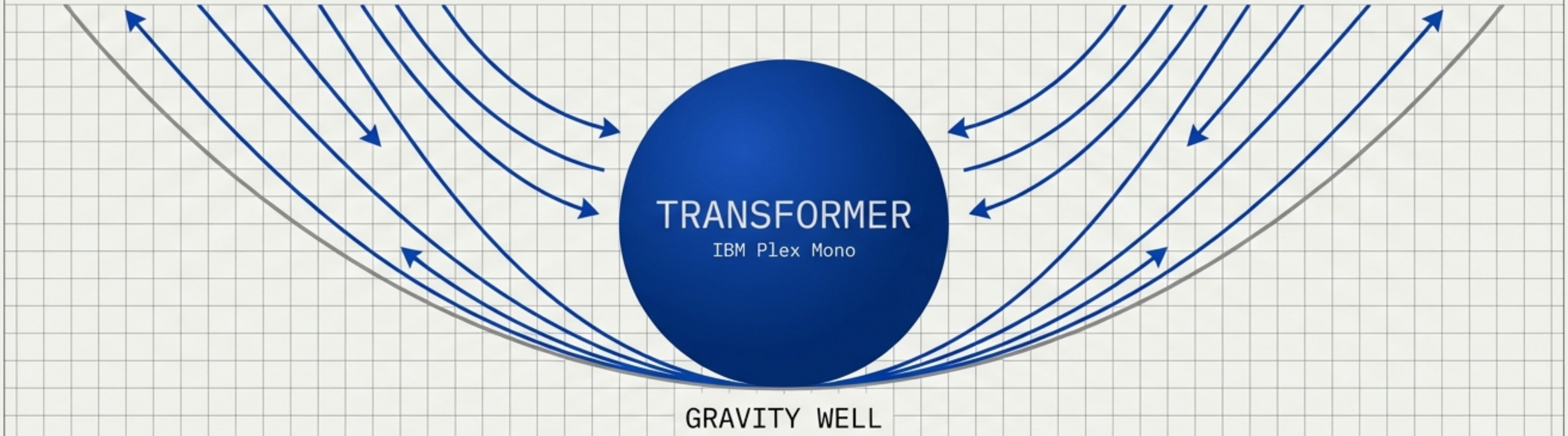
Avoid "AI Chip" bets unless there is a signed frontier lab partnership. Ecosystem inertia blocks the path.



SOURCE: ENGINEERING STRATEGY REPORT // DATE: 2028 // STATUS: CONFIDENTIAL / PRACTITIONER HEURISTICS

NotebookLM

The Transformer is the Stable Attractor.



We are in a self-reinforcing equilibrium bound by Hardware Specialization, Software Ecosystems, and Economic Incentives.

The constraints of the Transformer are no longer technical limitations; they are the geometry of the solution space.

NotebookLM

Sources & Further Reading

Mamba-2: Transformers are SSMs (Dao et al., 2024)

RetNet: A Successor to Transformer (Microsoft Research, 2023)

RWKV: Reinventing RNNs for the Transformer Era (Peng et al.)

NVIDIA Blackwell & Rubin Architecture Whitepapers

IBM Bamba Release Notes (2025)

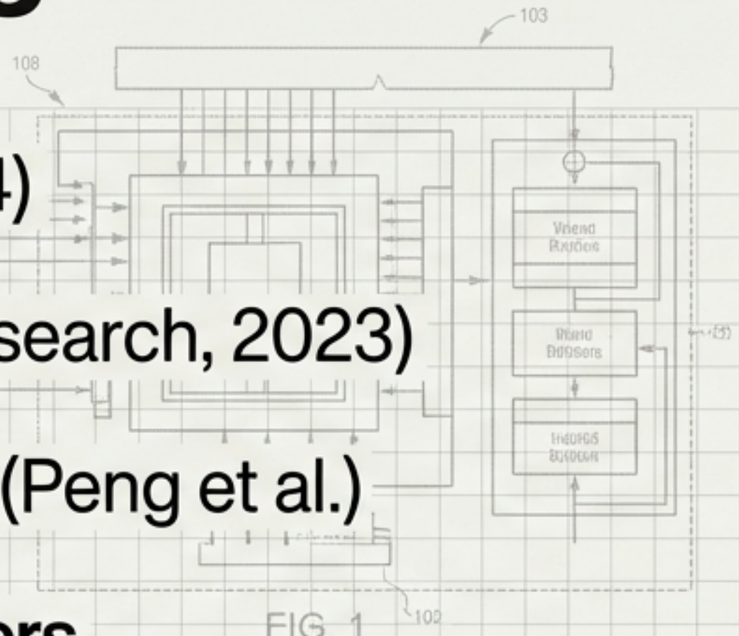


FIG. 1

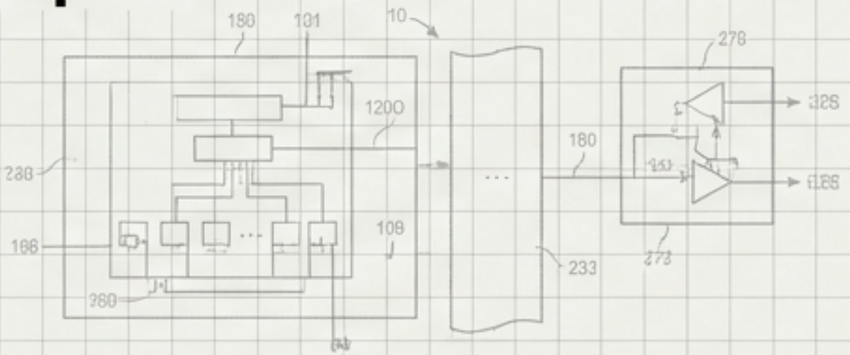


FIG. 4

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